

Answer Key — Biology Grade 11

Sub-Strand 3.2: Growth and Development in Animals • Lesson 1: The Empty Skin on the Maize Leaf:
Two Ways Insects Grow

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9
D	B	A	A	B	B	A	D	C

USE AFTER: PREDICT PHASE — Predict: drawing a before-and-after diagram of the exuvia beside the larger insect

Q1. A learner finds an empty, split insect skin on a maize leaf. A larger grasshopper is hanging beside it. Which statement describes what the learner OBSERVED?

- A. The insect peeled off its skin like sunburnt skin
- B. The skin stretched until it became the larger insect
- C. The insect melted and rebuilt itself inside the skin
- D. The empty skin is smaller than the insect beside it**

Correct answer: D. Only the first statement is something that can be seen directly. The other three are untested ideas (predictions) about how the insect got bigger, not observations.

USE AFTER: PREDICT PHASE — Predict: Think-Pair-Share and recording common ideas on the board

Q2. In the lesson, learners are asked to write a prediction about the empty skin. At this point, before the observation stations, which response best fits the teacher's instructions for ideas such as 'it peels like sunburn'?

- A. Tell the learner at once that the idea is wrong and give the correct explanation
- B. Record the idea in the learner's own words without correcting it, and test it later with evidence**
- C. Ignore the idea and move on to the next learner
- D. Remove the idea from the board because it is not scientific

Correct answer: B. The teacher records ideas in learners' own words and does not correct them, so later evidence creates a need to know.

USE AFTER: OBSERVE PHASE — Observe: Station 2 housefly larvae, pupae and adults, and sorting Sequence B

Q3. At the observation stations, learners study a housefly in its stages: maggot (larva), a motionless puparium (pupa) and an adult fly. Which stage sequence did they sort for the housefly?

- A. Egg, larva, pupa, adult**
- B. Egg, pupa, larva, adult
- C. Egg, nymph, nymph, adult
- D. Larva, egg, adult, pupa

Correct answer: A. The housefly follows complete metamorphosis with four stages in this order: egg, larva, pupa, adult.

USE AFTER: OBSERVE PHASE — Observe: Station 1 live grasshopper nymphs of different sizes plus an exuvia

Q4. Learners watch a young grasshopper nymph with a hand lens and compare it with an adult grasshopper. Which observation did the lesson lead them to record?

- A. The nymph looks like a small adult**
- B. The nymph has no legs, like a maggot
- C. The nymph is a motionless resting stage
- D. The nymph looks nothing like the adult

Correct answer: A. Grasshopper nymphs look like small adults, unlike housefly maggots, which look very different from adult flies.

USE AFTER: OBSERVE PHASE — Observe: sorting life cycle cards into Sequence A and Sequence B

Q5. A group sorts cards for a grasshopper and puts a pupa card between the nymph and adult cards. Which question should the teacher ask, as in the lesson?

- A. Why did you get this wrong?
- B. What evidence made you put this card here?**
- C. Who in the group made this mistake?
- D. Can you copy the answer from another group?

Correct answer: B. The teacher does not correct sorting errors directly but asks learners what evidence they used, so that they can recheck their own thinking.

USE AFTER: EXPLAIN PHASE — Explain: writing the first Claim-Evidence-Reasoning (CER) statement

Q6. A group has these observations: the housefly has a motionless pupa stage, but the grasshopper has no pupa stage. Which claim do these observations best support?

- A. All insects grow smoothly without changing shape
- B. Insects follow two developmental pathways**
- C. All insects have exactly four life stages
- D. Insects grow by peeling like sunburnt skin

Correct answer: B. One insect has a pupa (4 stages) and the other has none (3 stages). This supports two pathways: complete and incomplete metamorphosis.

USE AFTER: EXPLAIN PHASE — Explain: building the comparative model with grasshopper and housefly columns

Q7. How many stages are in the life cycle of a grasshopper (incomplete metamorphosis), and how many are in the life cycle of a housefly (complete metamorphosis)?

- A. Grasshopper 3 stages, housefly 4 stages**
- B. Grasshopper 4 stages, housefly 4 stages
- C. Grasshopper 3 stages, housefly 3 stages
- D. Grasshopper 4 stages, housefly 3 stages

Correct answer: A. Incomplete metamorphosis has egg, nymph, adult (3). Complete metamorphosis has egg, larva, pupa, adult (4).

USE AFTER: DQB PHASE — DQB: writing two questions on sticky notes and clustering them

Q8. Which question about the empty skin is the best kind of question to post on the Driving Question Board (DQB)?

- A. Is the skin on a maize leaf?
- B. Did the insect leave the skin?
- C. Is the skin crunchy?
- D. How does an insect become bigger than its old hard skin?**

Correct answer: D. A good DQB question starts with how or why and cannot be answered with yes or no.

USE AFTER: END OF LESSON — Model: bridging question and exit ticket

Q9. At the end of the lesson, which question is still NOT explained and is left for the next lessons?

- A. Which insect has a larva stage?
- B. How many stages does a grasshopper have?
- C. Why does a grasshopper grow in sudden jumps instead of a little each day?**
- D. Does a housefly have a pupa stage?

Correct answer: C. Learners have identified the stages and pathways, but the reason for growth in sudden jumps is the open question for Lesson 2.